

ROHIT KAPOOR

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OBJECTIVE

Final-year B.Tech (IT) student with hands-on experience in machine learning, deep learning, statistical testing, causal inference, and full-stack ML deployment. Seeking a Data Scientist / Data Analyst / ML Engineer / AI Engineer role.

EDUCATION

B.Tech in Information Technology — Jaypee University of Information Technology, Waknaghat 2022 – 2026
Senior Secondary (12th) — JCB Public Senior Secondary School, Shimla 2022

TECHNICAL SKILLS

ML & Statistics: XGBoost, CatBoost, Gradient Boosting, Random Forest, Logistic Regression, A/B Testing, Z-Test, T-Test, F-Test, ANOVA, Chi-Square, Cohen's h/d, Bonferroni Correction, Uplift Modeling, SHAP, Causal Inference, Power Analysis

Deep Learning & CV: CNN, RNN, LSTM, Transformers, ResNet, EfficientNet, Swin Transformer, PointPillars, Multi-task Learning, Transfer Learning, Grad-CAM, BEV Perception, Sensor Fusion

NLP & GenAI: Tokenization, Stemming (PorterStemmer, SnowballStemmer), Lemmatization, BERT, Text Classification, Sentiment Analysis, LangChain, RAG, ChromaDB, Prompt Engineering, Hugging Face

Data & Deployment: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, FastAPI, React, Streamlit, Docker, Git, SQL, Python, R

PROJECTS

Neural Drive — Multi-Task Learning for Autonomous Driving 2024

- Built a from-scratch multi-task perception system (35.7M params) fusing 6 surround cameras (Swin-T backbone + FPN + Lift-Splat-Shoot BEV projection) and LiDAR point clouds (PointPillars) via windowed cross-modal attention — reducing attention complexity from $O(HW^2)$ to $O(HW \cdot w^2)$ for memory-efficient fusion on 200×200 BEV grids.
- Simultaneously performs 3D object detection (CenterPoint heatmap head: cars, pedestrians, trucks) and HD map segmentation (14 classes); uncertainty weighting (Kendall et al.) auto-balances detection and segmentation losses. Trained on nuScenes Mini (Tesla T4, 24 epochs): train loss 2.11, val loss 2.31, detection heatmap confidence 0.993.

PyTorch · Swin-T · PointPillars · nuScenes · timm · einops · OpenCV · Python

A/B Test Analysis & Uplift Modeling — Email Marketing 2025

- Analyzed 64K-customer RCT using z-tests, Cohen's h effect sizes, and Bonferroni correction; Men's email +7.7pp lift ($p=5.69e-112$, $h=0.22$ — illustrating statistical ≠ practical significance).
- T-Learner meta-learner (Gradient Boosting) for individual treatment effects + SHAP + Qini curve evaluation; uplift targeting: 216% ROI vs 53% random — 4x more net value from same budget.

Python · XGBoost · SHAP · scikit-uplift · statsmodels · scipy · pandas · matplotlib

Brain Tumor MRI Classification with Grad-CAM — Full-Stack App 2025

- Fine-tuned ResNet18 (transfer learning) on 7,023 MRI images (4 classes); 95.75% test accuracy, 0.99 ROC-AUC. Grad-CAM revealed a source-distribution shortcut; intensity normalization + center-cropping suppressed it (94.7% → 95.75%). Full-stack FastAPI + React app with live heatmap.

PyTorch · ResNet18 · Grad-CAM · scikit-learn · FastAPI · Uvicorn · React · Vite · Python

RAG-Powered Document Assistant 2024

- PDF chatbot with vector retrieval (ChromaDB + sentence-transformers, top-6) and cross-encoder reranking (top-4); LLM-as-judge eval harness reduced hallucination rate 18% → 4%.

LangChain · ChromaDB · sentence-transformers · cross-encoder · FastAPI · React · Python

RESEARCH & PUBLICATIONS

Smart Manufacturing Dashboard for Industry 4.0 — ML-Based Predictive Maintenance & Supply Chain Risk 2025

IEEE Conference Paper · Jaypee University of Information Technology

- Isolation Forest anomaly detection + supervised supply chain risk classifier: 92.54% accuracy, 0.92 weighted F1. Cloud-deployed full-stack dashboard with RESTful APIs.

ACHIEVEMENTS & CERTIFICATIONS

- Top 100 globally — NeurIPS Polymer Prediction Challenge | Top ranks — Kaggle NeurIPS Math Misconception Competition
- AWS Foundations of Prompt Engineering (May 2026) | IBM SQL and Relational Databases 101 — CognitiveClass.ai (May 2026)